Experiment No.-

Aim: To obtain Three-phase to Two-phase conversion by Scott connection.

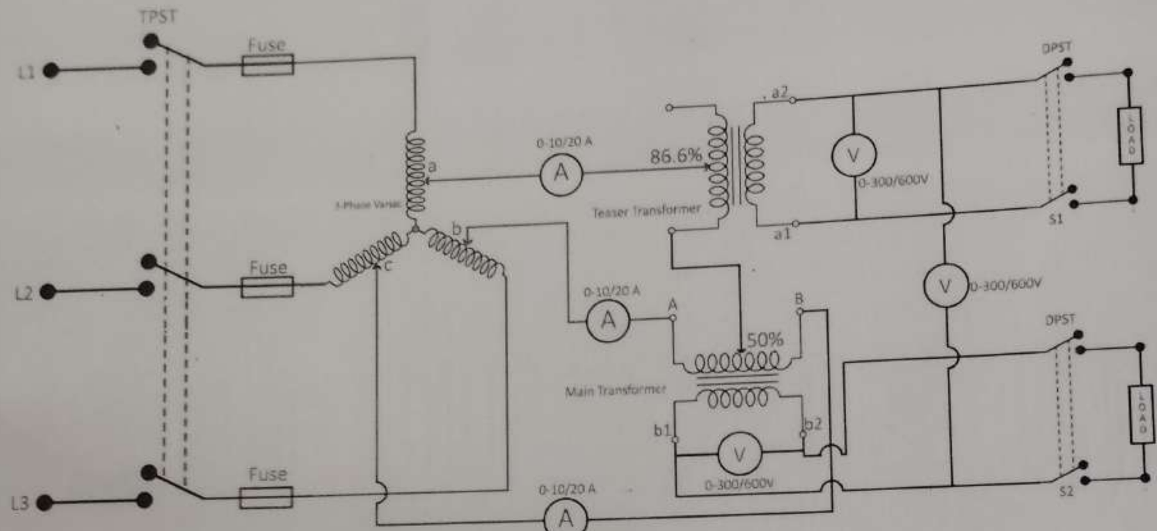
Instruments Required:

Sl.no.	Name	Range	Type	Quantity
1	3-Phase Variac	25A, 415/0-470V		1
2	Ammeter	0-10/20A	MI	3
3	Voltmeter	0-300/600V	MI	3

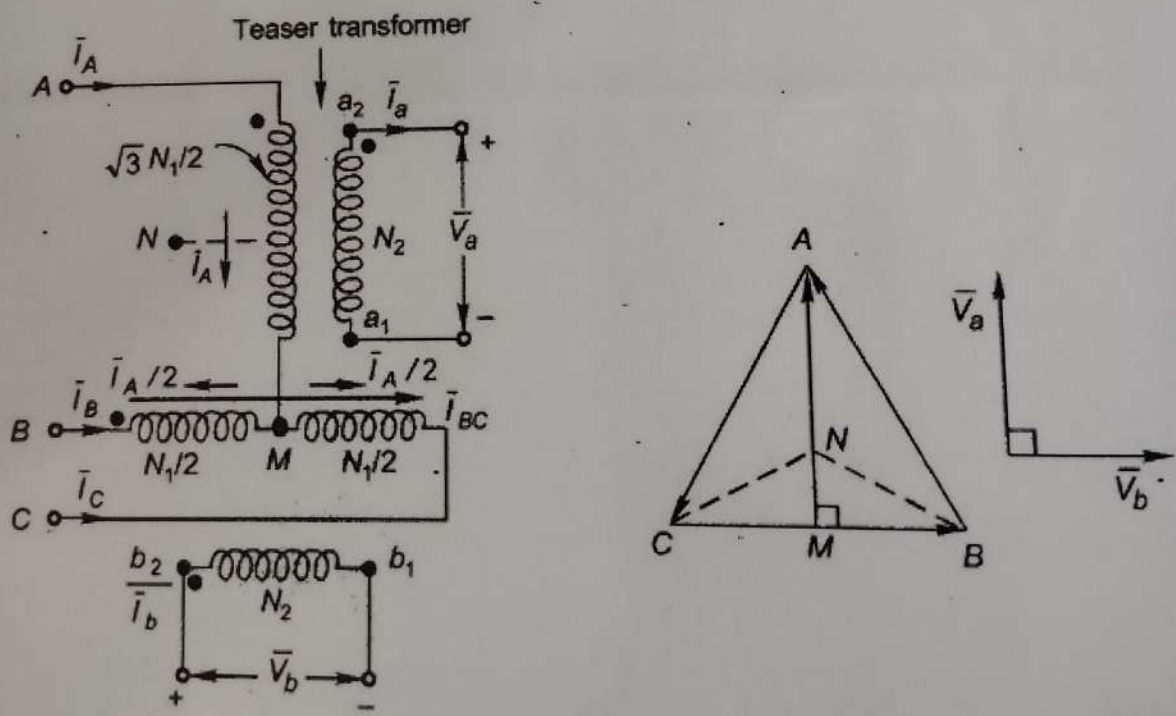
Theory:

Phase conversion from three to two phase is needed in special cases, such as in supplying 2-phase electric arc furnaces. The concept of 3/2-phase conversion follows from the voltage phasor diagram of balanced 3-phase supply shown in Fig. 3.61(b). If the point M midway on V_{BC} could be located, then V_{AM} leads V_{BC} by 90°. A 2-phase supply could thus be obtained by means of transformers; one connected across AM, called the **teaser transformer** and the other connected across the lines B and C. Since $V_{AM} = (\sqrt{3}/2)\ V_{BC}$, the transformer primaries must have $\sqrt{3}\ N1/2$ (teaser) and N1 turns; this would mean equal voltage/turn in each transformer.

A balanced 2-phase supply could then be easily obtained by having both secondaries with equal number of turns, N2. The point M is located midway on the primary of the transformer connected across the lines B and C. The connection of two such transformers, known as the Scott connection, is shown in Fig. 3.61, while the phasor diagram of the 2-phase supply on the secondary side is shown in Fig. The neutral point on the 3-phase side, if required, could be located at the point N which divides the primary winding of the tertiary in the ratio 1:2 (refer Fig.).



Circuit Diagram



(a) (b) (c)

Procedure:

1. Connections are done as per the circuit diagram.
2. By using 3-Phase auto transformer apply different voltages to the circuit.
3. Observe different meter readings and note them in observation table.

Observation Table:

	3-Ø Supply			2-Ø Supply		
S.no	V_a(volt)	V_b(volt)	V_c(volt)	V_a(volt)	V_b(volt)	V_c(volt)
1						
2						
3						
4						
5						

Result:

Precautions:

1. Do not touch any terminal or switch without ensuring that it is dead.
2. Make sure that all the connections are right, before switching on any circuit, faulty connections may cause drawing large power resulting in damage of parts of instruments.
3. The circuit should be de-energized before changing any connections.